Q1: What does the difference in loss between the Ridge model and the baseline indicate? Specifically, when comparing MAE and business loss, on which metric does the Ridge model perform better? Based on these results, can we conclude that the Ridge model produces better predictions than the baseline_weekday_mean method? Why or why not?

Across the two backup datasets, the difference in loss shows that Ridge is optimizing the business objective rather than purely minimizing prediction error. In the high-demand scenario (backup_data1), Ridge improves both MAE (2803.4 vs 2947.3) and business loss (74358 vs 78528), indicating clear overall superiority. In the sparse-demand scenario (backup_data2), however, Ridge has a worse MAE (49.36 vs 44.80) but a slightly lower business loss (1203 vs 1206), meaning its predictions are less accurate on average but still reduce cost.

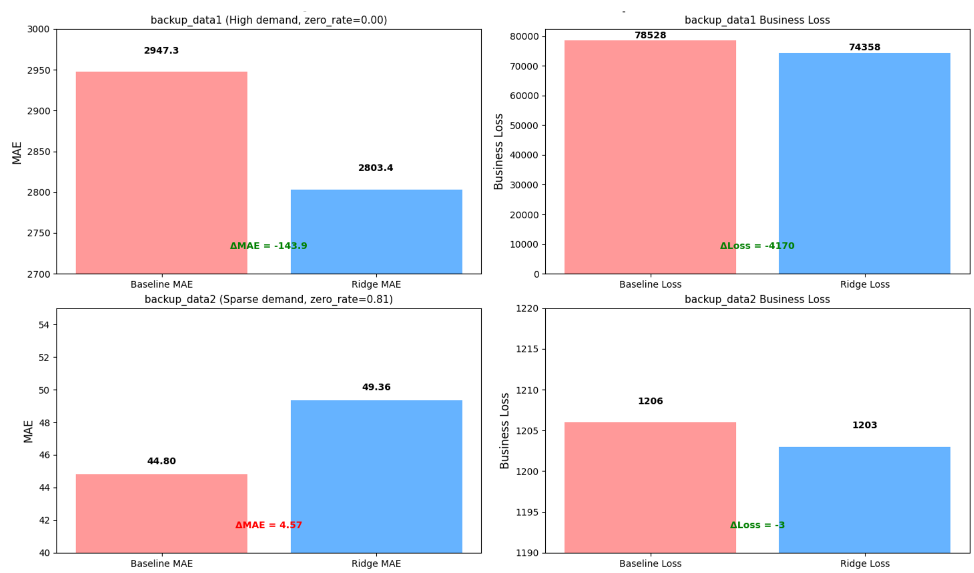

Figure 1

This demonstrates that Ridge performs better on business loss, not consistently on MAE. The reason is that the loss function penalizes underestimation more heavily than overestimation, and Ridge is better at capturing high-demand cases, thereby avoiding costly understaffing errors—even if that increases average prediction error.

The actual dataset (Figure 2) supports this pattern: Ridge again shows higher MAE but slightly lower loss, confirming that its advantage mainly lies in cost-sensitive performance rather than accuracy.

Therefore, we cannot conclude that Ridge universally produces better predictions than the baseline_weekday_mean method. Ridge is clearly better in high-demand settings and slightly better in terms of business loss overall, but it sacrifices MAE in some cases. Its superiority depends on the objective: it is preferable for minimizing asymmetric business cost, but not for maximizing overall predictive accuracy.

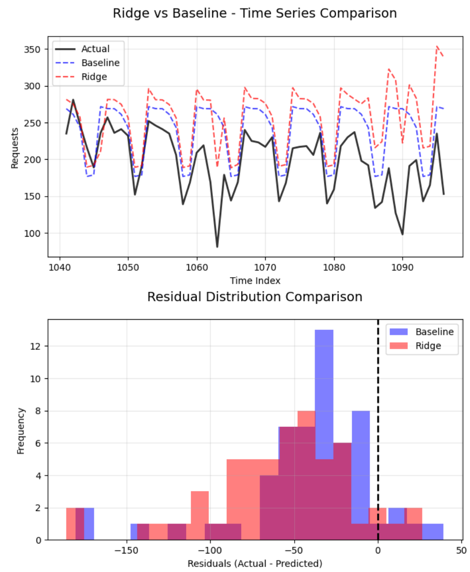

Figure 2

Q2: Check both slices and answer following question: Is smoothing always help reduce loss? Why or why not?

Smoothing does not always reduce loss across the three datasets. It helps only when it aligns with the underlying structure of the data and can hurt when it removes important signals.

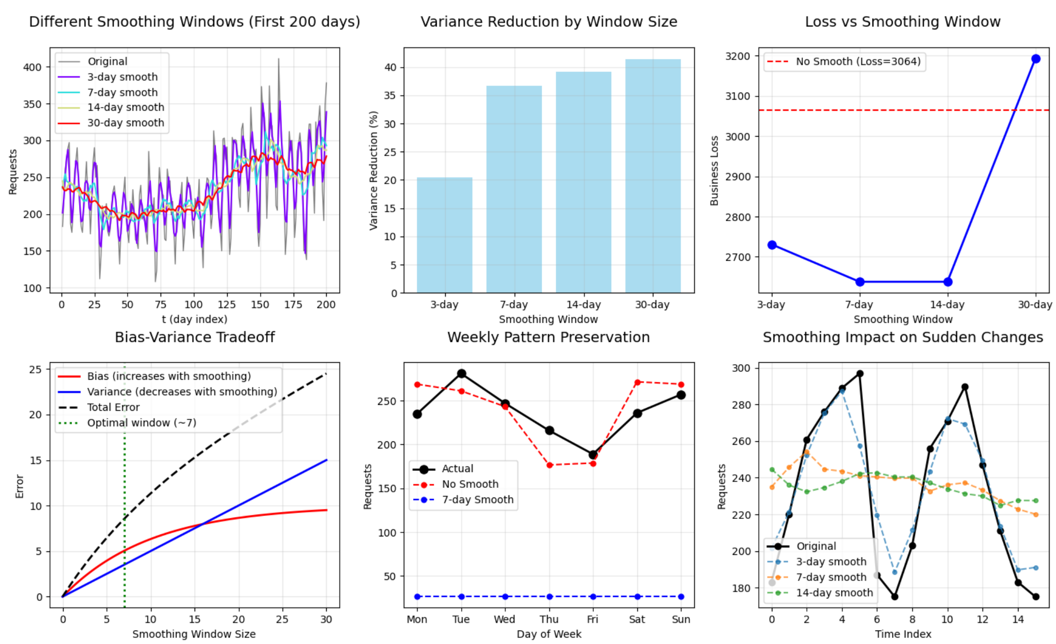

Figure 3

For the actual dataset (Figure 3), smoothing improves performance up to a point: a 7-day window gives the lowest loss because it matches the natural weekly pattern, balancing noise reduction and signal preservation. However, larger windows (14-day, 30-day) increase loss because they oversmooth, washing out meaningful variations such as weekday effects or sudden spikes.

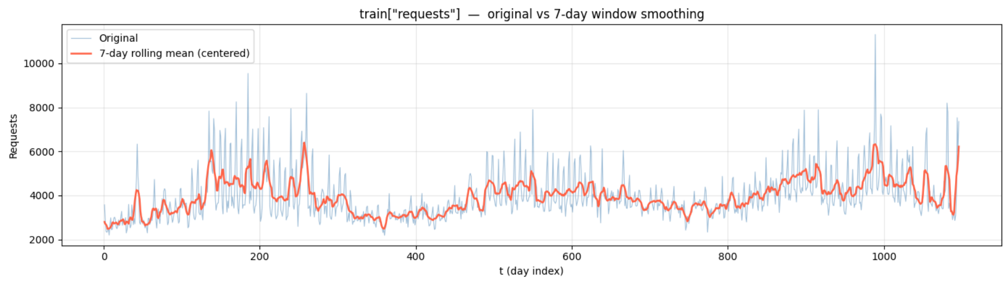

Figure 4 Backup data 1

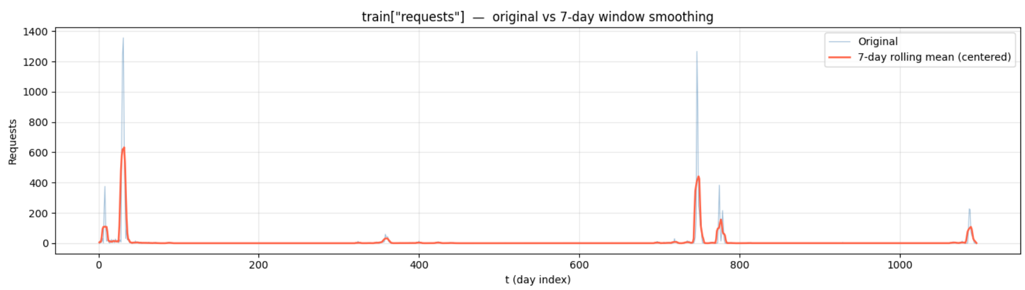

Figure 5 Backup data 2

The two backup datasets show why this happens more clearly. In the first backup (high-volume, noisy but structured series), moderate smoothing reduces variance and clarifies trends, similar to the real data. In contrast, the second backup (mostly sparse with sharp spikes) demonstrates that smoothing can be harmful: it spreads and dampens rare but important spikes, distorting the signal and potentially increasing prediction error.

Overall, smoothing reduces loss only when noise dominates and patterns are regular. It fails—or even increases loss—when the data contains sharp changes, sparse events, or when the smoothing window is too large, because this introduces bias and removes critical information.

Forecasting & Staffing Optimization — NYC 311 Demand

1. Problem statement

This project aims to forecast NYC 311 daily service requests and translate predictions into staffing decisions under an asymmetric business cost. Understaffing is penalized twice as much as overstaffing, making accurate peak prediction critical. Therefore, model selection is based mostly on business loss, ensuring operational relevance.

2. Data Exploration & Feature Selection

The training set covers 1,096 days (mean 236, std 61 req/day). EDA revealed two dominant seasonal patterns: a weekly cycle (weekdays 260 vs weekends 177, a 47% premium) and an annual cycle (July peak 281 vs December trough 184). Figure 1 shows that is_weekend carries the strongest negative correlation with requests (r = −0.62), while early-week dummy variables (dow_0 to dow_2) are the strongest positive predictors (r ~0.20). month and week_of_year are near-collinear (r = 0.97); both are therefore encoded as sin/cos pairs rather than raw integers to avoid redundancy and preserve cyclic adjacency.

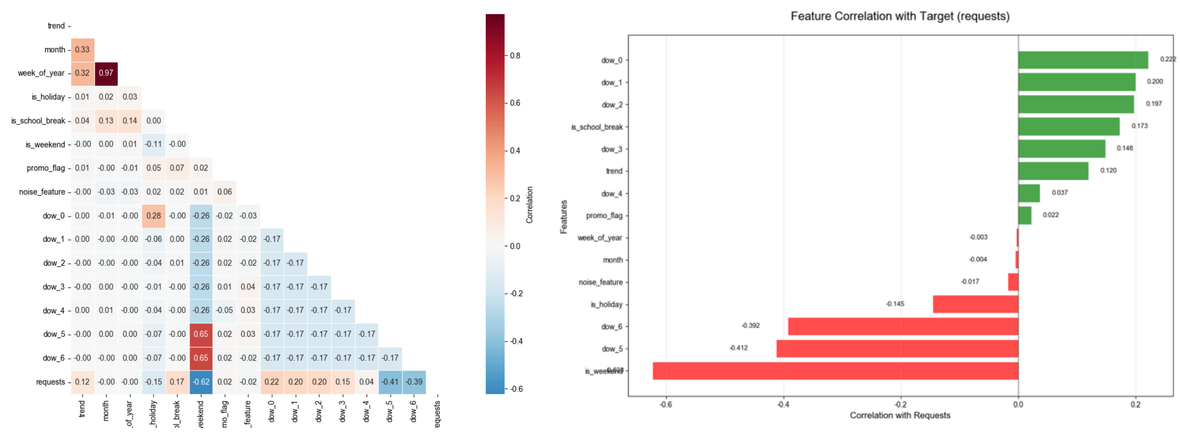

Figure 6 Feature Correlation Plot

3. Model Design - HistGradientBoosting

Ridge achieved lower loss than baseline but higher MAE, confirming the need for a cost-sensitive objective. HistGradientBoosting outperformed both in MAE and loss.

The choice of HistGradientBoosting over Ridge was driven by three factors:

(a) Non-linearity: Ridge assumes linear relationships, but EDA revealed interactions (e.g., holiday × weekday, weekend × month) that linear models cannot capture. Tree-based models handle these naturally.

(b) Lag feature handling: Ridge requires all features to be numeric and non-missing, making it difficult to use NaN-generating lag features (e.g., lag_7, lag_14). HistGradientBoosting natively handles missing values, allowing direct inclusion of lag features without imputation.

(c) Feature heterogeneity: The feature set includes binary flags, cyclical encodings (sin/cos), and rolling statistics. Ridge's linear structure struggles with such mixed scales; tree models are scale-invariant and automatically select relevant splits.

Lag features are the most important design decision. The autocorrelation at lag-7 is r=0.763 — the highest of any feature — confirming that "same weekday last week" is the strongest signal in the data. Lags 14, 21, and 28 also peak strongly (r=0.75, 0.73, 0.70), showing a persistent weekly memory structure.

This motivated the core lag feature:
same_dow_ewm = 0.5×lag7 + 0.3×lag14 + 0.15×lag21 + 0.05×lag28

This recency-weighted blend of the same weekday across four weeks reduces sensitivity to single anomalous weeks (e.g., a holiday on the prior Monday inflating the next Monday's prediction).

Additional rolling features:
trend7vs14 = roll7 − roll14: captures demand momentum; critical for detecting the Nov→Dec seasonal descent
level_ratio = roll7 / roll28: anchors recent level relative to the medium-term mean

Bias Correction:
The post-prediction multiplier bias_mult=0.94 was tuned via grid search (0.80–1.00). Values below 0.92 increased overstaffing days excessively; values above 0.96 increased understaffing. 0.94 balanced the two, minimizing total business loss.

Because staffing is determined by ceil(pred/30), a 6% reduction often pushes a prediction from just above a multiple of 30 to just below it, avoiding an extra staff unit. This reduces wasted capacity in low-demand months while rarely causing understaffing, since the original over-prediction was systematic.

4. Results

Model	MAE	Business Loss	MAE Improvement	Loss Improvement
Weekday-mean baseline	44.07	3064	-	-
HistGBR with lags	20.05	1573	-54.5%	-48.7%

Figure 7

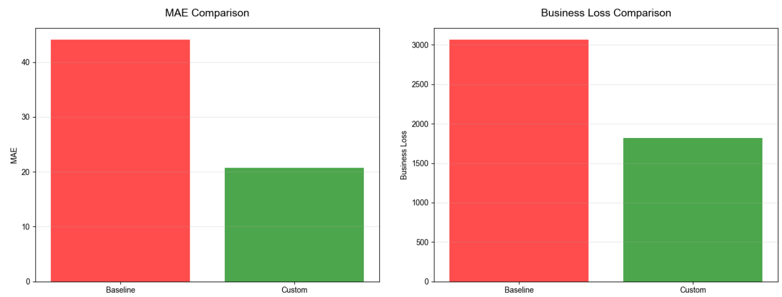

Figure 8 MAE & Business Loss Comparison

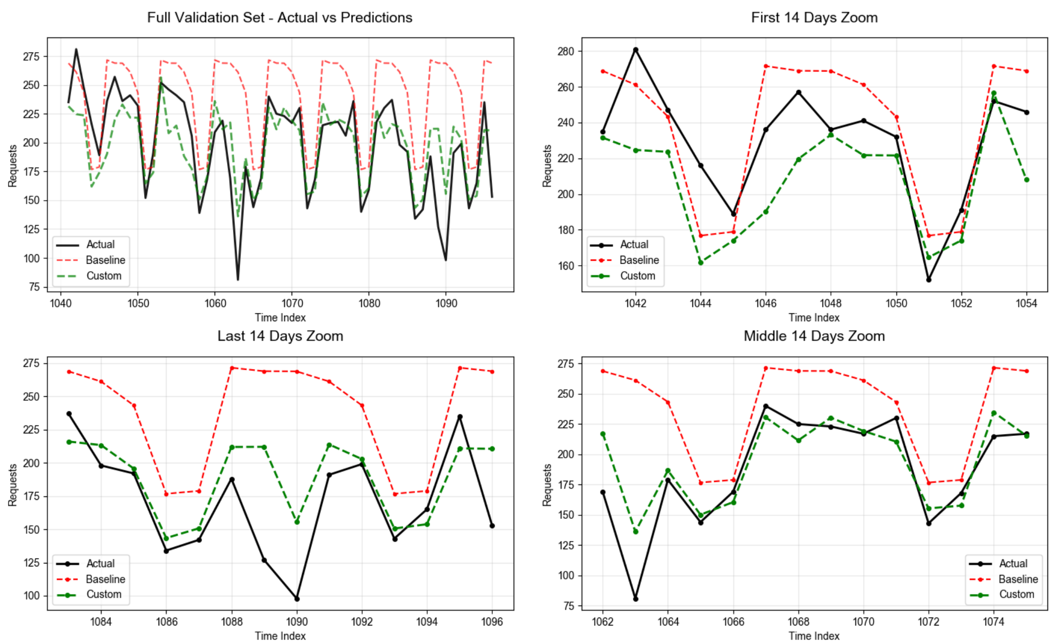

Figure 9 Validation Set

From figure 9 we can see validation time series (days 1041–1096, Nov–Dec). The custom model (green) tracks the seasonal demand descent, while the baseline (red) over-predicts throughout, wasting staffing budget.

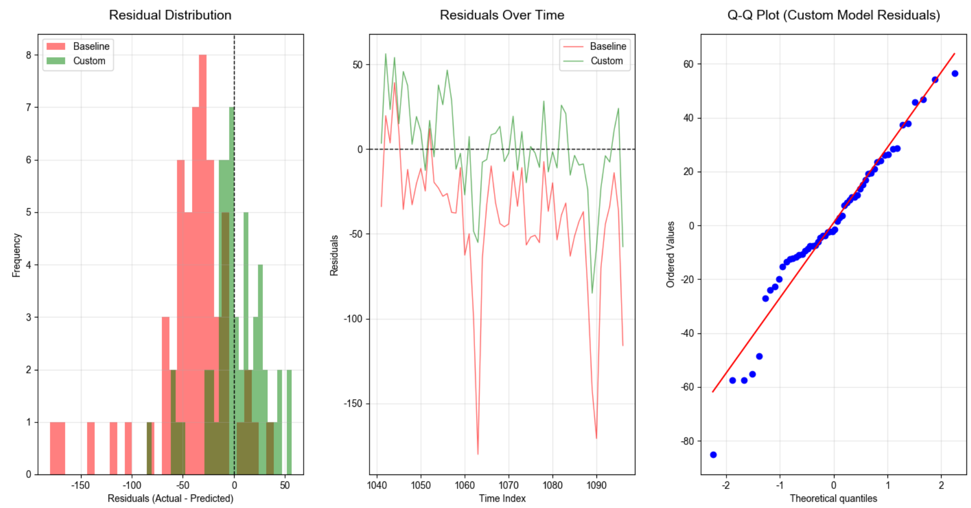

Figure 10

From an error perspective, the custom model significantly reduces both the magnitude and variability of residuals. The distribution of residuals is more centered and less skewed, indicating improved calibration. In contrast, the baseline model shows a wider spread with more extreme negative errors, reflecting frequent overprediction. The time-series residual plot further confirms this, with the custom model exhibiting fewer large spikes and more stable performance over time.

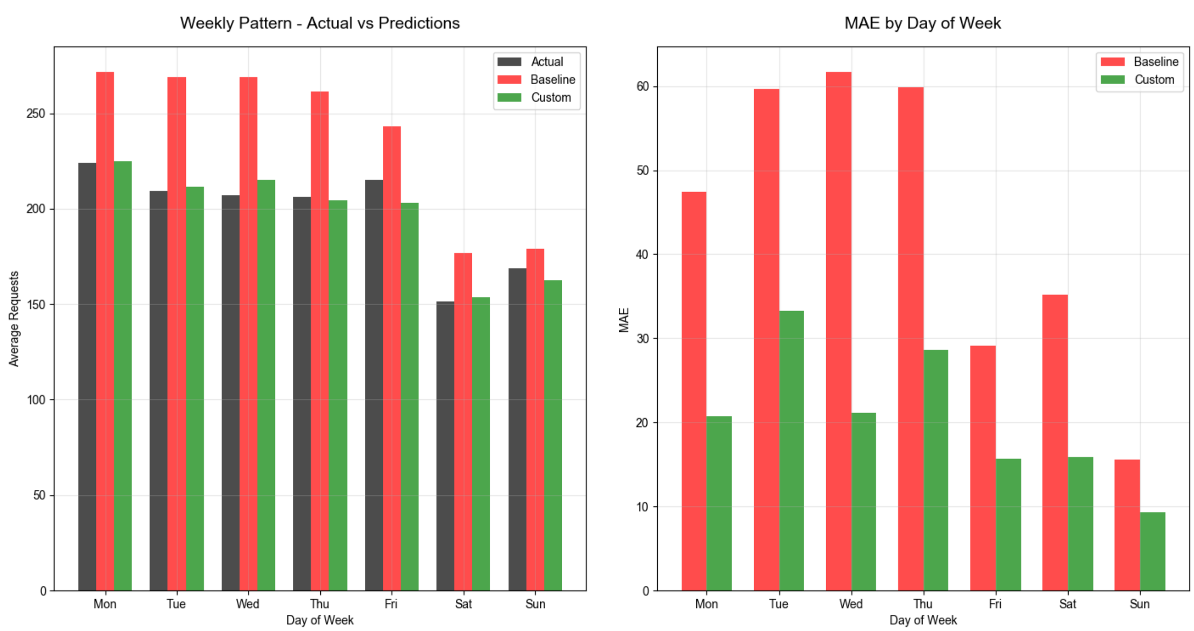

Figure 11

The custom model reduces error on every day; weekdays (Mon–Thu) see the largest absolute improvement because the baseline over-predicts most severely on those days.

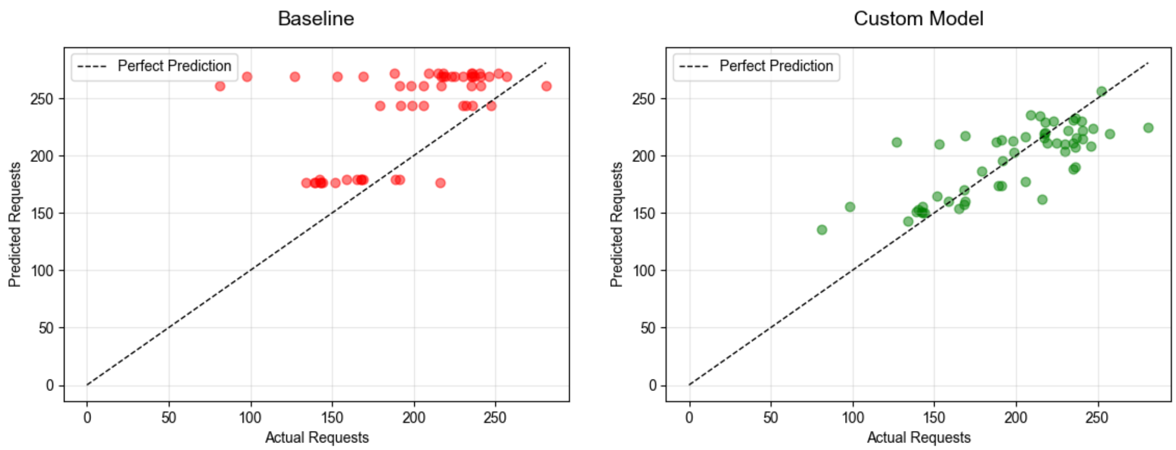

Figure 12

The baseline clusters horizontally at fixed levels while the custom model aligns tightly to the y=x diagonal.

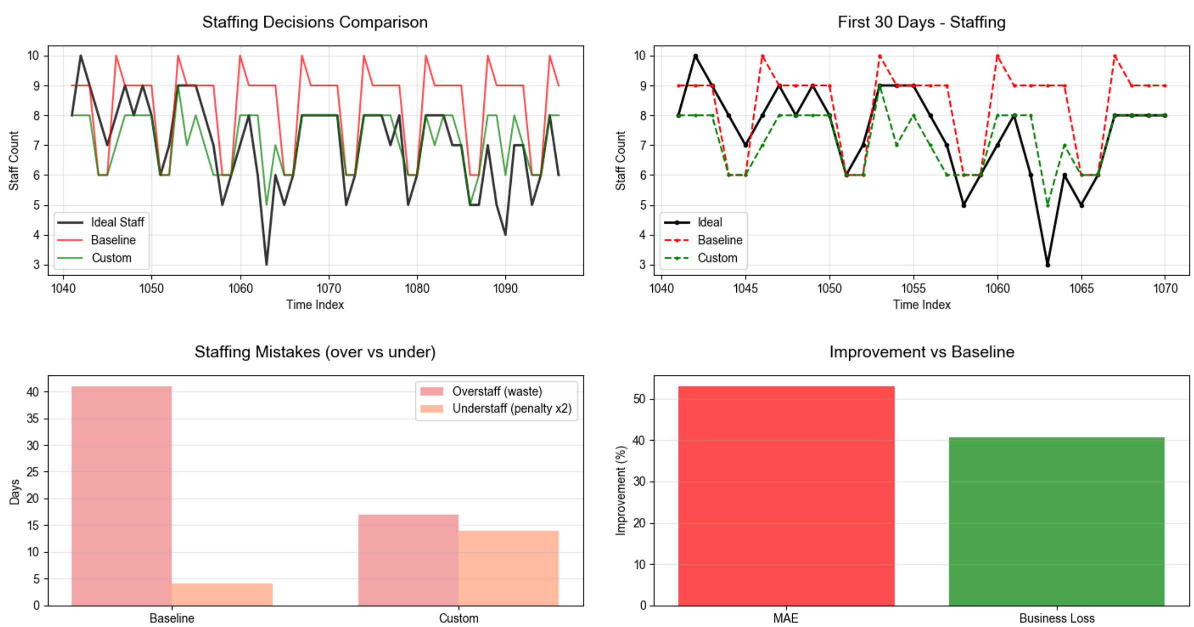

Figure 13

The custom staffing model improves both accuracy and business outcomes compared to the baseline. It follows the ideal staffing levels more closely and adapts better to demand changes, particularly in the first 30 days, avoiding the baseline's consistent overstaffing.

Operationally, it achieves a better balance between overstaffing and understaffing. While slightly increasing understaffing, it significantly reduces excess staffing, leading to a more efficient overall trade-off and net gain.

5. Interpretation & limitations

The baseline repeats a fixed weekday-average template regardless of season, causing severe over-staffing in the Nov–Dec window (262 predicted vs 197 actual). The custom model tracks the seasonal descent by leveraging rolling-window momentum features. Also, the predictions of custom model align closely to actuals across the full demand range, while the baseline collapses to two horizontal bands.

However, no external regressors (weather, events) are included. Extreme anomaly days still produce large residuals because same-weekday lags do not anticipate sudden demand shocks. The 0.94 bias multiplier is tuned on a single 56-day validation window; a cross-validated correction would generalise more reliably across seasons.

6. Conclusion

The proposed model significantly improves both forecast accuracy and operational cost efficiency. By capturing seasonality, trends, and nonlinear interactions, it reduces unnecessary staffing while maintaining service levels.